



Effects of cluster training sets on muscle power and force–velocity relationship in postmenopausal women

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Abstract

Purpose The purpose of this study was to investigate whether the higher load resistance training with a cluster training sets (CS) design maximizes muscle power and strength gains when compared to higher load resistance training with a traditional sets (TS) design in postmenopausal women (PW).

Methods Each leg of 16 PW was randomly allocated into two groups: TS ($n = 16$ legs) and CS ($n = 16$ legs). Both groups performed three sets of four repetitions at 90% one repetition maximum (1RM), 3 s per muscle action, with a 1.5-min rest interval between sets, twice-weekly, for 8 weeks. Only CS group performed 30 s interrepetition rest periods.

Results Both groups similarly increased ($P < 0.05$) thigh muscle mass, muscle strength (1RM), and maximal muscle power. However, whereas the CS increased ($P < 0.05$) peak power at 40% and theoretical maximal velocity (V_0), the TS increase theoretical maximal force (F_0). The TS reduced slope of the force–velocity relationship when compared to the CS.

Conclusions Although both CS and TS design are similarly effective at improving muscle mass and strength and $P_{\max}$, TS and CS induce different changes in the force–velocity profile (S_{FV}) in PW. Thus, our findings suggest that TS may be a preferential RT design if the focus of training is a force–velocity profile more “force-oriented”, whereas CS may be a preferential RT design if the focus of training is a force–velocity profile more “velocity-oriented”.

Keywords Interrepetition rest · Hypertrophy · Muscle strength · Older women

Introduction

It is well evidenced that muscle function (strength and power) progressively decline with aging [1]. A decline in muscle function lead to functional limitation and disability [2], which contribute to increased risks of falling [2, 3], hospitalization [4] and mortality [4, 5] in older people.

Specifically, older women have lower muscle function when compared to older men [6]. As the postmenopausal period is accompanied by an accentuated decline in muscle function [6–9], women show a more rapid decline in physical function when compared to men during middle age [6, 8, 9]. Due to increased life expectancy, women spend one-third of their lives in a postmenopausal period. Consequently, the impact of impairments in muscle function on health care expenditure is superior in older women than to older men [10]. Thus, improving muscle function in older people may have a positive impact on prevention of development of functional limitation and disability [3, 11], extending their healthspan [12], especially in postmenopausal women (PW).

Resistance training (RT) is well accepted as an intervention to improve muscle strength and power in older people [3, 11, 13]. Different configurations of RT variables provoke different acute physiological and mechanical responses, leading to different muscular adaptations [3, 11, 13]. Hence, it is assumed that RT variable manipulations (i.e., load and introduce short interrepetition rest periods

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between repetitions) may maximize muscular adaptations [3, 11, 13, 14]. However, the most appropriate RT program design to maximize muscle strength and power in older adults remains elusive.

Higher load RT [$\geq 70\%$ of maximal muscle strength (1RM)], performing repetitions until, or close to, volitional fatigue without interrepetition rest periods (8–12 repetitions) and multiple sets (i.e., three sets) have been widely recommended as a RT program design to maximize muscle strength gains in older adults [13, 15]. However, the configuration of a set in higher load RT (i.e., no interrepetition rest periods) results in an acute decline of movement force and velocity and also of technical proficiency from first to last repetition (i.e., the length of set) [16–18], probably due to increased metabolic demand and neuromuscular fatigue [17–19]. Moreover, the decline in these parameters is increased in subsequent sets [17, 18]. Recently, it has been shown that intraset fatigue applied chronically during an RT program may lead to the reduction of the myosin heavy chain IIX and negatively influences muscle power gains [20–22]. As muscle power is the product of force and velocity of muscle shortening, a decline in muscle power requires a change in force, velocity or both [1]. Moreover, a higher number of sets (higher volume), which may lead to higher fatigue [13], mitigate muscle power gains in older adults [11, 23]. Thus, the RT program design recommended to maximize muscle strength gains (i.e., higher load RT and no interrepetition rest periods) may not be the RT program design ideal to maximize muscle power gains, especially gains in velocity component of muscle power [20].

The introduction of short interrepetition rest periods (traditionally defined as cluster training sets (CS) [14]) has shown avoid intraset fatigue in young people [24]. Moreover, CS result in a less acute reduction of movement force and velocity and also of technical proficiency from first to last repetition (i.e., set length) when compared to a traditional set (TS) design in young people [25]. Consequently, CS has shown to maximize muscle strength and power gains when compared to a traditional RT in young people [26]. However, the body of evidence regarding acute and chronic responses of CS on muscle strength and power in older adults is extremely scarce. To the best of our knowledge, no study has evaluated the influence of a higher load RT with a CS design on muscle strength and power in older people. Hence, the purpose of this study was to investigate whether the higher load RT with a CS design maximizes muscle power and strength gains when compared to higher load RT with a TS design in PW. We hypothesized that higher load RT with a CS design maximizes muscle power gains when compared to higher load RT with a TS design in PW.

Methods

Experimental approach to the problem

Sixteen PW were selected for this study. The women's legs were randomized to one of two RT protocols: TS (16 legs) and CS (16 legs). We used a tool of Medcalc® (Create random group) to assign cases to random groups (randomize cases). However, in the TS group, one woman reported pain in one knee and did not perform all the tests at the baseline. Thus, the TS group was composed of 15 legs.

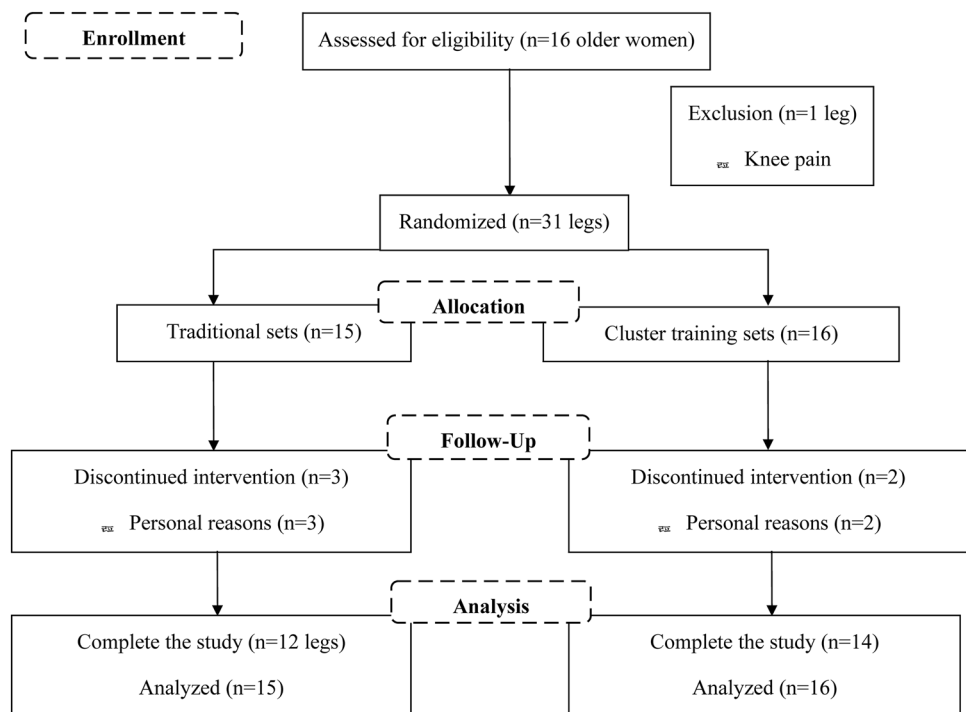
This randomized study was conducted over 10 weeks. Initially, it was performed 1 week of familiarization of RT (2 sessions) in both legs. After this, anthropometric measurements and all baseline tests (maximal muscle strength and power measurements after 20 min of maximal muscle strength) were performed in 1 week (1 session). Then, both legs performed 8 weeks of RT at unilateral leg extension machine (isotonic exercise), with a frequency (no consecutive days) of two times per week. During the training, one women's leg performed the TS protocol, while the other leg of the same women performed the CS protocol in the same session of training, one at a time. The training order of the legs was changed at each training session. Finally, anthropometric measurements and all tests were performed 72-h after the last session of training to avoid residual effects of training. Three women discontinued the intervention (five legs) and did not perform the evaluations of the study end. However, as randomized controlled trials suffer from missing outcomes, we included every subject who was randomized according to randomized treatment assignment (Fig. 1).

Subjects

All PW (age 63.3 ± 6.7 years, body weight 69.9 ± 16.7 kg, height 1.6 ± 0.1 cm and body mass index 28.6 ± 5.6 kg/m²) reported being housewives and not performing physical exercise. All PW were between 52 and 74 years old whose amenorrhea had occurred at least 12 months before the study. The inclusion criteria consisted of: no hormone therapy or phytoestrogens; controlled blood pressure and glycaemia; absence of myopathies, arthropathies, and neuropathies; absence of muscle, thromboembolic and gastrointestinal disorders; absence of cardiovascular and infection diseases; non-drinker (no alcohol intake whatsoever in their diet), non-smoker and no participation in a regular RT program in the previous 6 months. A CONSORT diagram is shown in Fig. 1.

All the volunteers were clear on the objectives and procedures of the study and gave us their written informed

Fig. 1 CONSORT flow diagram allocation the volunteers in traditional sets group and cluster training sets group



consent. The study was approved by the University Review Board for the Use of Human Subjects (no. 2.654.326, approved by the local Ethics Committee) and was written in accordance with the standards set by the Declaration of Helsinki.

Anthropometric assessments

The body weight and height were measured with a digital scale (Lider®, Brazil) and a stadiometer fixed to the scale, respectively. Body mass index ($BMI = \text{weight}/\text{height}^2$) was classified according to the system used by the World Health Organization. The thigh skinfolds were evaluated with the caliper (Lange®) on both sides of the body by an experienced examiner. The thigh circumference was measured in the right thigh with an inextensible tape with 200 cm of length and precision of 1 mm. The anthropometric equation for estimation of thigh cross-sectional area (CSA) was $(\text{thigh circumference} - \pi \times \text{skinfold})^2/4\pi$ [27].

Maximal muscle strength

Before the one repetition maximal (1RM) test, all volunteers (both legs) participated a 1-week familiarization period (2 sessions) with low loads, on alternating days, to become familiar with the exercise technique (unilateral leg extension). The evaluation of 1RM was performed in both legs, at two times during the study: test pre-training and again after 8 weeks of training. The 1RM performance was quantified as the maximal weight lifted in unilateral

leg extension machine (Physical®, Rio de Janeiro, RJ, Brazil). The procedures of the 1RM test were performed as described by Nunes et al. [28]. PW performances in the 1RM test (kg) were used as a muscle strength indicator in TS and CS groups.

Power, force and velocity measurements

Power, force and velocity were measured in both legs during a unilateral leg extension (from 90° to ~10° of knee flexion) at a leg extension machine with the hip set at 90°. Peak power (PP), peak force (PF) and peak velocity (PV) were measured during each repetition using a dynamometer isoinertial (linear encoder), the Peak Power® (Cefise, Nova Odessa, SP, Brazil). Only the data of the concentric phase were used. This dynamometer isoinertial is interfaced to Peak Power 4.0 software (Cefise, Nova Odessa, SP, Brazil). The configurations and calibration followed the manufacturer's specifications. The dynamometer isoinertial was attached to the cable of leg extension machine (near of weight stack) to register the time and the displacement of movement at a frequency of 30 Hz.

The subjects performed one set of five continuous repetitions at 40%, 50% and 60% of 1RM after 20 min of the 1RM test. Between percentual of 1RM (40%, 50%, and 60%), a rest of 1.5 min was allowed. All volunteers were encouraged to perform each repetition as fast and strongly as possible. We recorded the highest peak power value (consequently, PF and PV) of the five continuous repetitions.

Force intercept, velocity intercept, force–velocity slope, and maximal muscle power analyses

Individual force–velocity relationship was obtained from the highest peak power recorded for every load of the 1RM test (40%, 50%, and 60%). Individual regression linear was obtained considering force and velocity at the highest peak power value [29–32]. Force at zero velocity (i.e., theoretical maximal force isometric force) was obtained from the force intercept (F_0), while the velocity at zero force (i.e., maximal velocity without load) was calculated as the velocity intercept (V_0). The slope of the force–velocity relationship (S_{FV}) was obtained from the equation: $-F_0/V_0$. The maximal muscle power ($P_{\max}$) was calculated from the equation: $(F_0 \times V_0)/4$ [29, 32].

Resistance training protocols

The RT protocols follow the recommendations of the American College of Sports Medicine (ACSM) Guidelines for optimizing gains in maximal muscle strength [13]. All training sessions were performed in a gym at the local University. All volunteers performed twice-weekly (no consecutive days) routine for 8 weeks and a fitness professional supervised all training sessions. Before each training session, the volunteers performed a warm-up (walking during 5 min at a usual speed). The TS performed three sets of four repetitions (without interrepetition rest periods) at 90% 1RM, 3 s per muscle action, with a 1.5-min rest interval between sets. The CS performed three sets of four repetitions (30 s interrepetition rest periods) at 90% 1RM, 3 s per muscle action, with a 1.5-min rest interval between sets. To ensure a load at 90% of 1RM for four repetitions during eight weeks of training, the training loads were adjusted in the 4th week.

Statistical analyses

All analyses were performed using SPSS software (version 23.0, IBM Corp, New York, NY, USA). Changes in all variables (Tables 1, 2, 3) were analyzed by an intention-to-treat analysis using Generalized Estimating Equations and using unstructured correlation and maximal likelihood estimation. A low significance difference (LSD) post hoc was used to a pairwise comparison. The data were presented as means and Wald 95% confidence interval. The effect size with pooled standard deviation was used to compare the deltas (Δ) between groups. Cohen provided benchmarks to define small ($d = 0.2$ – 0.49), medium ($d = 0.5$ – 0.79), and large ($d \geq 0.80$) effects. The significant level was set at $P < 0.05$.

Table 1 Thigh cross-sectional area and maximal muscle strength following 8 weeks of traditional sets vs. cluster training sets in postmenopausal women

	TS ($n = 15$)		CS ($n = 16$)		Δ	P group	P time	P time vs. group	d
	Pre	Post	Pre	Post					
CSA (cm^2)	225.7 [193.9 to 262.8]	242.3 [203.6 to 288.9]	16.8 [−0.8 to 34.3]	202.2 [183.9 to 222.4]	216.3 [196.7 to 237.8]	14.1 [6.4 to 21.7]	0.235	<0.001	0.912
1RM (kg)	30.4 [25.0 to 36.8]	34.0 [28.2 to 41.1]	3.6 [2.7 to 4.6]	30.5 [25.2 to 37.0]	33.1 [28.5 to 38.4]	2.6 [0.9 to 4.3]	0.928	<0.001	0.352

All values are means and 95% confidence interval. Generalized estimating equation test

TS traditional sets, CS cluster training sets, CSA thigh cross-sectional area, 1RM maximal muscle strength in unilateral leg extension, d d_{Cohen} with a correction of a positive bias in the pooled standard deviation

Table 2 Peak power, force and velocity at 40%, 50% and 60% of 1RM measurements following 8 weeks of traditional sets vs. cluster training sets in postmenopausal women

	TS (<i>n</i> = 15)			CS (<i>n</i> = 16)			<i>P</i> group	<i>P</i> time	<i>P</i> time vs. group	<i>d</i>
	Pre	Post	Δ	Pre	Post	Δ				
40% of 1RM										
PP (w)	235.2 [171.1 to 323.2]	250.8 [176.9 to 355.7]	15.7 [−2.8 to 34.1]	252.5 [171.3 to 372.2]	321.7 ^a [201.8 to 512.8]	69.2 [129.8 to 8.5]	0.561	0.001	0.019	0.58
PF (N)	182.6 [150.3 to 221.9]	210.8 [165.1 to 269.1]	28.2 [6.9 to 49.5]	186.1 [144.0 to 240.6]	222.8 [161.3–307.7]	36.7 [5.6–67.8]	0.838	<0.001	0.624	–
PV (m/s)	1.24 [1.08–1.42]	1.14 ^a [1.00–1.30]	−0.10 [−0.20–0.00]	1.26 [1.13–1.41]	1.30 [1.16–1.47]	0.05 [−0.01–0.10]	0.383	0.301	0.014	0.95
50% of 1RM										
PP (w)	234.9 [180.1–306.2]	263.8 [186.7–372.7]	28.9 [−7.2–65.0]	268.9 [182.9–395.5]	322.9 [204.3–510.5]	54.0 [2.2–105.8]	0.522	<0.001	0.430	–
PF (N)	213.1 [174.4–260.4]	253.5 [196.6–330.3]	40.4 [12.7–68.1]	224.0 [169.5–296.0]	242.6 [183.7–320.2]	18.6 [−9.7–46.8]	0.998	<0.001	0.210	–
PV (m/s)	1.04 [0.95–1.14]	1.00 [0.91–1.11]	−0.04 [−0.00–0.08]	1.13 [1.03–1.25]	1.16 [1.01–1.34]	0.03 [−0.05–0.10]	0.130	0.809	0.107	–
60% of 1RM to										
PP (w)	223.2 [174.2 to 286.0]	264.1 [189.4 to 368.3]	40.9 [0.3 to 81.5]	272.4 [189.1 to 392.2]	316.5 [206.2 to 485.7]	44.1 [−0.5 to 88.8]	0.445	<0.001	0.835	–
PF (N)	235.1 [194.5 to 284.2]	282.5 [225.7 to 353.6]	47.4 [23.5 to 71.2]	265.0 [211.8 to 331.6]	303.9 [232.9 to 396.4]	38.8 [8.8 to 68.9]	0.552	<0.001	0.396	to
PV (m/s)	0.97 [0.87 to 1.08]	0.89 [0.79 to 0.99]	−0.08 [−0.20 to 0.04]	0.97 [0.86 to 1.09]	0.96 [0.84 to 1.10]	−0.01 [−0.06 to 0.05]	0.606	0.188	0.278	–

All values are means and 95% confidence interval. Generalized estimating equation test

TS traditional sets, CS cluster training sets, PP peak power, PF peak force, PV peak velocity, *d* d_{Cohen} with a correction of a positive bias in the pooled standard deviation

^aDifference from pre-value (LSD post hoc test)

Table 3 Theoretical maximal isometric force, maximal velocity without load, maximal muscle power and slope of force–velocity relationship measurements following 8 weeks of traditional sets vs. cluster training sets in postmenopausal women

	TS (<i>n</i> = 15)			CS (<i>n</i> = 16)			<i>P</i> group			<i>P</i> time vs. group			<i>d</i>
	Pre	Post	Δ	Pre	Post	Δ	Pre	Post	Δ	Pre	Post	Δ	
F_0 (N)	426.6 [355.5 to 511.9]	581.2 ^a [454.1 to 744.0]	154.7 [27.1 to 282.2]	524.1 [418.5 to 656.5]	506.1 [388.7 to 659.1]	– 18.0 [– 92.6 to 56.6]	0.824	0.048	0.013	0.84	0.013	0.002	0.84
V_0 (m/s)	2.56 [1.97 to 3.32]	2.11 [1.72 to 2.59]	– 0.45 [– 1.06 to 0.15]	2.03 [1.78 to 2.32]	2.68 ^a [2.07 to 3.46]	0.64 [0.13 to 1.14]	0.980	0.607	0.002	0.99	0.002	0.002	0.99
$P_{\max}$ (w)	256.4 [189.6 to 346.7]	288.5 [206.9 to 402.3]	32.1 [– 3.4 to 67.6]	286.1 [200.1 to 408.9]	362.9 [226.8 to 580.9]	76.9 [2.2 to 151.5]	0.520	< 0.001	0.206	to	0.206	0.001	to
S_{fv}	– 207.8 [– 154.5 to 280.4]	– 287.7 [– 202.0 to 384.8]	– 76.9 [– 188.9 to 34.9]	– 261.0 [– 220.3 to 307.4]	– 202.5 [– 156.2 to 260.3]	58.5 [– 3.4 to 120.3]	0.726	0.885	0.033	0.90	0.033	0.003	0.90

All values are means and 95% confidence interval. Generalized estimating equation test

TS traditional sets, CS cluster training sets, F_0 theoretical maximal isometric force, V_0 maximal velocity without load, $P_{\max}$ maximal muscle power, S_{fv} slope of the force–velocity relationship, d Cohen's d with a correction of a positive bias in the pooled standard deviation

^aDifference from pre-value (LSD post hoc test)

Results

Table 1 shows the CSA and 1RM measurements before and after 8 weeks of interventions. 1RM performance and CSA, when analyzing TS and CS, showed significant effect of training (time effect, $P < 0.001$), despite no group effect.

Table 2 shows values of PP, PF, and PV at 40%, 50% and 60% of 1RM before and after 8 weeks of interventions. PP and PF at 50% and 60% 1RM, when analyzing TS and CS, showed significant effect of training (time effect, $P < 0.001$), despite no group effect. The PV did not change at 50% and 60% of 1RM after the interventions. PF at 40% of 1RM, when analyzing TS and CS groups, showed significant effect of training (time effect, $P < 0.001$), despite no group effect. However, there was a statistical effect of group for PV at 40% of 1RM (significant time vs. group effects, $P = 0.014$), indicating that PV decreased in the TS group (LSD post hoc: $P < 0.05$). Also, the PP at 40% of 1 RM was increased in the CS, but not in TS (significant time vs. group effects, $P = 0.019$ and LSD post hoc: $P < 0.05$).

Table 3 shows $P_{\max}$, F_0 , V_0 and S_{fv} changes in both groups. $P_{\max}$, when analyzing TS and CS groups, showed significant effect of training (time effect, $P < 0.001$), despite no group effect. However, significant interactions of time vs. groups ($P < 0.05$) were observed in F_0 and V_0 , indicating that the TS group increased F_0 , whereas the CS groups increased V_0 (LSD post hoc: $P < 0.05$). There was a significant interaction of time by groups to S_{fv} , ($P = 0.033$), indicating that the force–velocity profile was more “force-oriented” (i.e., a steeper slope) in the TS, whereas in the CS the profile was more “velocity-oriented” (Fig. 2a, b).

Discussion

To the best of our knowledge, this randomized study was the first to compare the effects of high-load RT with TS and CS on muscle mass, strength and power and force–velocity relationship in PW. We hypothesized that higher load RT with a CS design maximizes muscle power gains when compared to higher load RT with a TS design in PW. Our results showed that although TS and CS were similarly effective to improve CSA, 1RM, $P_{\max}$, and PP at higher external resistance (50% and 60% of 1RM), CS was superior to TS for improving PP at lower external resistance (40% of 1RM), confirming, at least in part, our hypothesis. Moreover, we found that TS was effective for improving theoretical maximal isometric force (i.e., F_0), whereas CS was effective for improving maximal velocity without load (i.e., V_0), indicating that TS and CS induced different changes in the force–velocity relationship (i.e., S_{fv}) in PW.

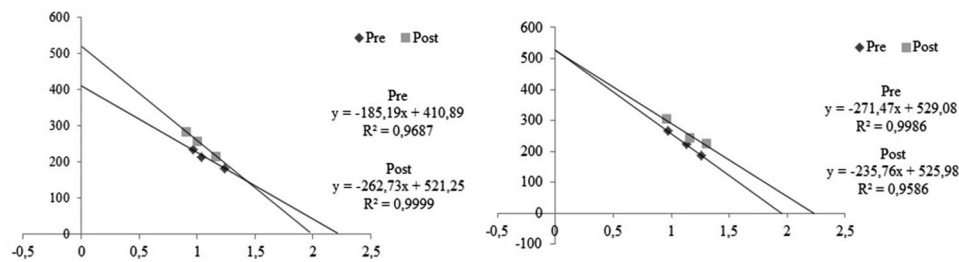


Fig. 2 Effects of the traditional sets and cluster training sets on the slope of force–velocity relationship. Y axis: force intercept. X axis: velocity intercept. **a** Values of the slope of force–velocity relationship pre and post training for theoretical maximal isometric force and

maximal velocity without load in the traditional sets group. **b** Values of the slope of force–velocity relationship pre and post training for theoretical maximal isometric force and maximal velocity without load in the cluster training sets group

PW have lower muscle mass and strength when compared to older men [6, 33]. Low muscle strength and mass are well-known risk factors for low functional capacity (i.e., walking, chair stand, and balance) in older women [2, 34]. Thus, improving muscle strength and mass in older women may prevent the development of disability [2, 34]. In this sense, our results show that CS improves muscle mass (CSA) and strength (1RM) in older women similar to TS and previous interventions with traditional high-load resistance training in older women [35, 36]. Thus, muscle mass and strength gains seem to occur independently of interrepetition rest period. These results are in line with previous studies that reported there was no difference between TS and CS designs in muscle strength and mass gains in young people [26, 37]. Although TS and CS are similar in muscle strength and mass gains, short interrepetition rest periods may avoid intraset fatigue and maintain peak force across a set of repetition [24, 38, 39]. Consequently, CS seems to be especially important for older women who do not tolerate fatigue during RT.

It has been shown that muscle power declines at an earlier and faster rate with aging than muscle strength [2, 40]. As muscle power is more strongly associated with functional capacity than muscle strength [41, 42], muscle power has been suggested as an important therapeutic target for an intervention aimed at preventing disability in later life [2]. In the current study, TS and CS similarly improved the $P_{\max}$ and PP at higher load of 1RM (50% and 60%) in older women. These results corroborate the findings of a great deal of the previous studies which found higher load RT increases $P_{\max}$ and PP at different percentages of 1RM [1, 11, 13, 20, 23]. However, a particular novelty of the current study is the greater increase in PP at lower load of 1RM (40%) with the CS when compared to the TS (27.4% vs. 6.6%, respectively). This finding has important implications, because muscle power at lower external resistance (i.e., 40% of 1RM) seems to explain more of the variability in capacity functional than muscle power at higher external resistance (i.e., 70% of 1RM) [43]. Collectively, these findings corroborate the findings of Ramirez-Campillo et al. who also

found that CS induces significantly greater improvements in functional capacity than the traditional RT set design in older women [44].

Muscle power is the product of force and movement velocity. Hence, an increase in muscle power requires a change in force, velocity or both [45]. In the current study, the increased muscle power at 40%, 50% and 60% of 1RM were due to an increased PF in both groups. However, the greater increase in muscle power at 40% of 1RM in the CS group was also due to a difference in PV between groups. Whereas the TS group significantly reduced the PV by 8.0%, the CS increased (not significant) by 3.2%. The traditional RT set design provokes a higher metabolic demand and a decline in capacity performance (velocity, force, and power) across the set [17–19, 24] when compared to CS [24, 46], which may result in different adaptations, such as reduced gains in velocity, strength and power [20–22]. Recently, it has been shown that this intraset fatigue may reduce myosin heavy chain IIX [20–22]. Myosin heavy chain IIX is the fastest and powerful isoform [21]; therefore, a reduction in this isoform might be detrimental to velocity [22]. Thus, CS is an RT effective to attenuate velocity loss during RT session [46] and may be used to optimize velocity gains (at lower resistances) in PW.

In the current study, we observed that determining the maximal muscle power or power solely at a percentage of maximal strength capacity (i.e., 60% and 50% of 1 RM) may not identify specific adaptation concerning to the design of resistance exercise programs. The force–velocity relationship has recently been suggested to be a more advantageous approach than determining the muscle power at a single percentage of maximal strength (i.e., % of 1RM) [32, 47, 48]. However, few studies have investigated the role of RT on a force–velocity relationship in older people [1]. In the current study, TS increased maximal force (F_0) gains, whereas CS increased maximal velocity gains (V_0) in older women. Our findings suggest that TS induces a force–velocity profile (S_{fv}) more “force-oriented” and CT induces a force–velocity profile (S_{fv}) more

“velocity-oriented” in PW. Our findings are important, since older adults may exhibit different force–velocity profile [1, 47], it is possible to guide a more adequate training depending on the force–velocity profile of the older adult.

This study is not without limitations. First, a limitation of the study is related to a small number of participants due to the nature of the intervention, which may have weakened the statistical analyses. Second, a short duration of RT program (8 weeks). It has been accepted that muscle adaptations occur during the early stages of training and this does not guarantee that such muscle adaptations will be similarly sustained in all groups after 8 weeks. This is an important issue for future research (> 8 weeks). Third, our PW are apparently healthy. Thus, our results cannot be generalized to other populations, such as women with frail or sarcopenia. Finally, the unilateral leg extension exercise employed in the present study may promote a contralateral effect (cross education), which may lead to induced muscle strength gains in contralateral limb [49]. Thus, strength gains in one leg may have been influenced by the cross-education effect of trained contralateral limb, promoting similar results between legs (i.e., between groups) in 1RM and PF at 50% and 60%. However, the cross-education effect seems to have been minimal on F_0 , V_0 , S_{fv} and PF at 40% gains. Moreover, it has been shown no correlation between legs (left and right) for strength gains after different RT protocols [50]. This suggests that when both limbs are trained, the cross-education effect is minimal [50].

Conclusions

Although both CS and TS designs are similarly effective at improving muscle mass, strength (1RM), peak power at 50% and 60% of 1RM and P_{max} , CS increases peak power at lower resistance, whereas TS increased F_0 (theoretical maximal force) indicating that TS and CS induce different changes in the force–velocity profile (S_{fv}) in PW. Thus, our findings suggest that if a force–velocity profile more “force-oriented” is the focus of training, TS may be a preferential RT design. On the other hand, if a force–velocity profile more “velocity-oriented” is the focus of training, CS may be a preferential RT design. Moreover, CS may be an important strategy for improving muscle power in lower resistance and for PW who do not tolerate fatigue during RT.

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Compliance with ethical standards

Conflict of interest The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical approval The study was given by the University Review Board for the Use of Human Subjects (no. 2.654.326, approved by the local Ethics Committee) and was written in accordance with the standards set by the Declaration of Helsinki.

Informed consent Written informed consent was obtained from all individual women included in the study.

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